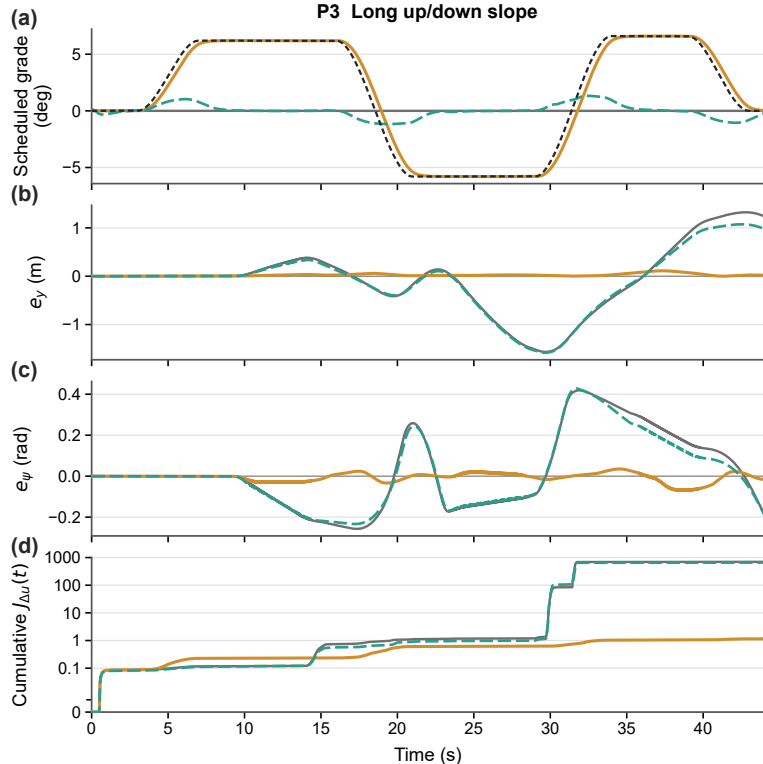


----- True grade — Zero-grade — Truth-driven reference - - Causal-IMU



P3 traces are descriptive; route-bootstrap inference uses all six routes. The cumulative $J_{\Delta u}(t)$ axis uses a symmetric-log scale and its endpoint equals the frozen route metric.

(e)

Six-route relative outcomes

	e_y RMS	e_ϕ RMS	$J_{\Delta u}$
P1	T ↓ 99.9% I ↓ 0.7%	T ↓ 99.6% I ↑ 27.1%	T ↓ 100% I ↑ 49.9%
P2	T ↓ 98.4% I ↓ 25.9%	T ↓ 91.7% I ↓ 21.9%	T ↓ 100% I ↓ 56.9%
P3	T ↓ 94.8% I ↓ 5.9%	T ↓ 87.7% I ↓ 6.7%	T ↓ 99.8% I ↓ 6.0%
P4	T ↓ 4.4% I ↑ 19.0%	T ↓ 5.2% I ↑ 3.6%	T ↑ 97.3% I ↑ 8.4%
P5	T -- I ↑ 4.9%	T -- I ↑ 7.3%	T -- I ↓ 5.9%
P6	T ↓ 88.4% I ↓ 0.4%	T ↓ 95.7% I ↓ 4.3%	T ↑ 105% I ↓ 2.6%
95% CI	T: benefit I: benefit	T: benefit I: crosses 0	T: benefit I: crosses 0

T = truth-driven; I = causal-IMU; ↓ lower/better; ↑ higher/worse

Percentages use each route's zero-grade value as the baseline.